
REQUIREMENTS NOT MET

- N/A

VIDEO FILE LINK

https://youtube.com/shorts/_I65pF8lryo?feature=share

PROBLEMS ENCOUNTERED

- Understanding how to derive inputs from outputs, took a bit!

FUTURE WORK/APPLICATIONS

We can now create counters starting from any position using state machines, creating a feedback loop, this could be used to create arithmetic logic units (ALUs) in the future.

PRE-LAB QUESTIONS OR EXERCISES

<insert copy of pre-lab exercises/questions from lab document, as well as an answer directly following each of the exercises/questions (if not applicable, write "N/A")>

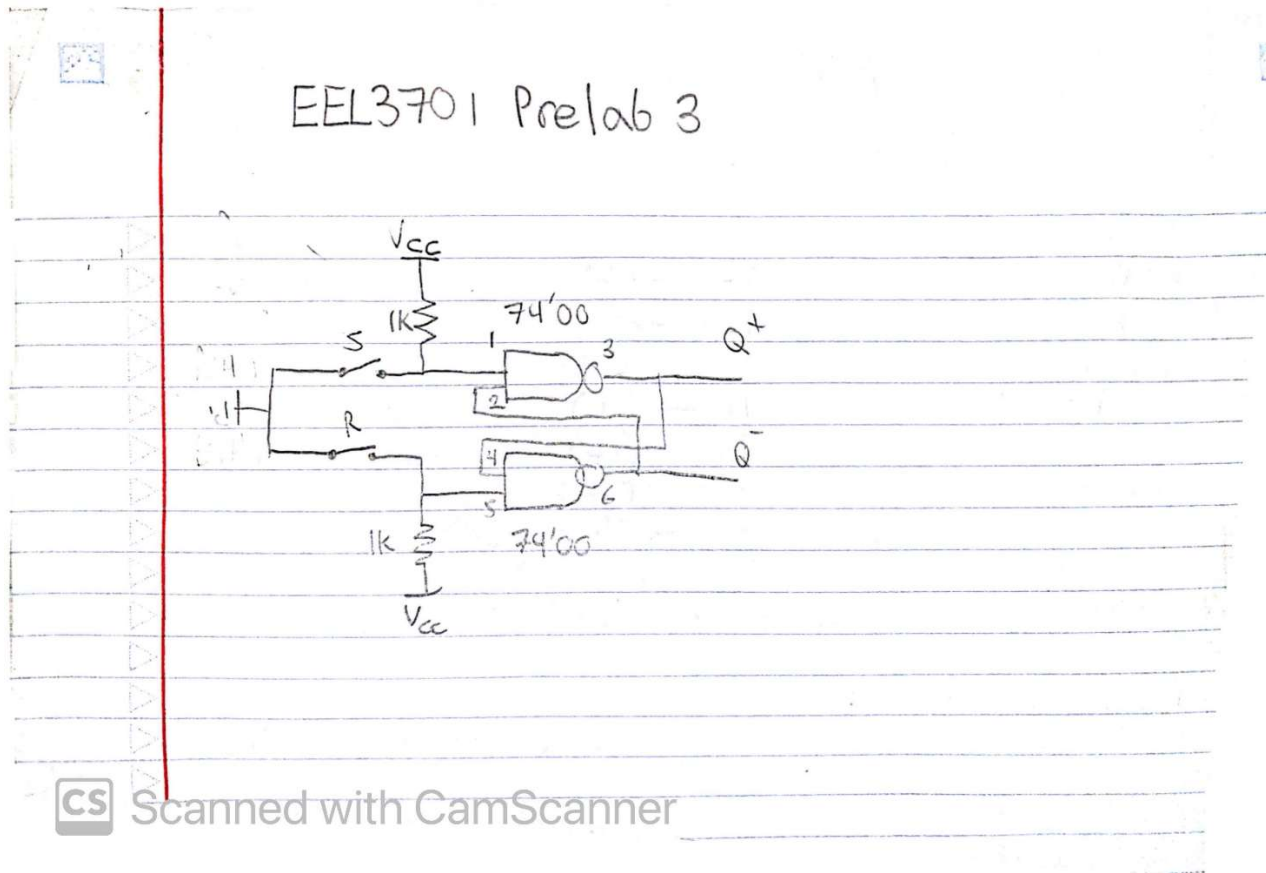


Figure 1: Proposed Debounce Switch Circuit Pinout



Figure 2: Non-debounced Switching Response

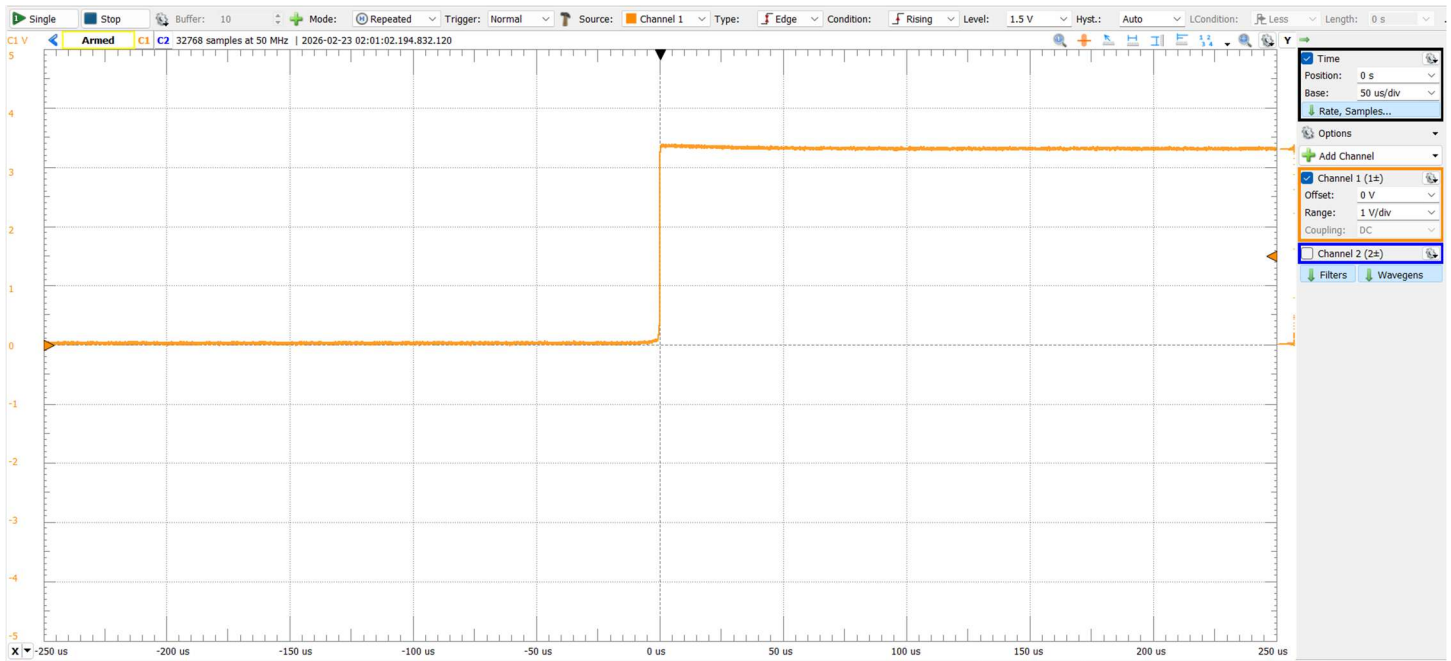


Figure 3: Non-debounced Switching Response 2

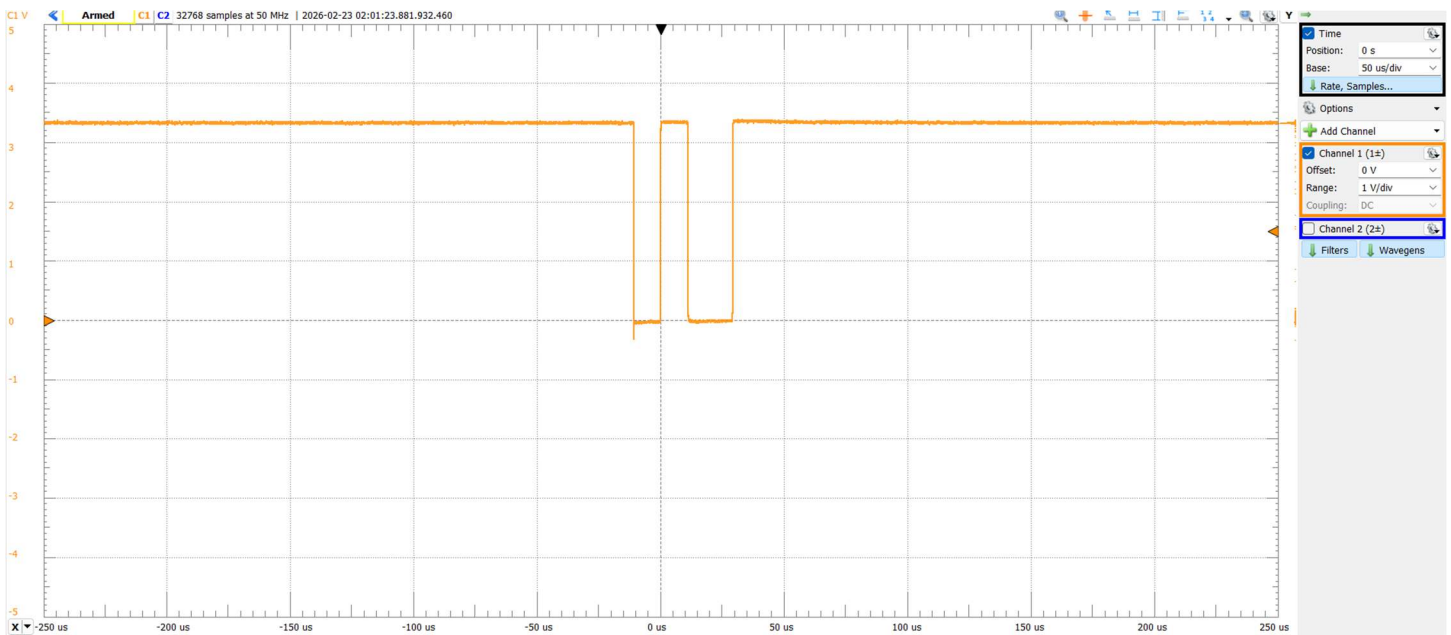


Figure 4: Non-debounced Switching Response 3

Since the clock looks for a rising edge, some of these would cause errors, the first and third image would cause at least two switches per, and are likely to fail more often.

Debounced after this point



Figure 5: Debounced Switch Response

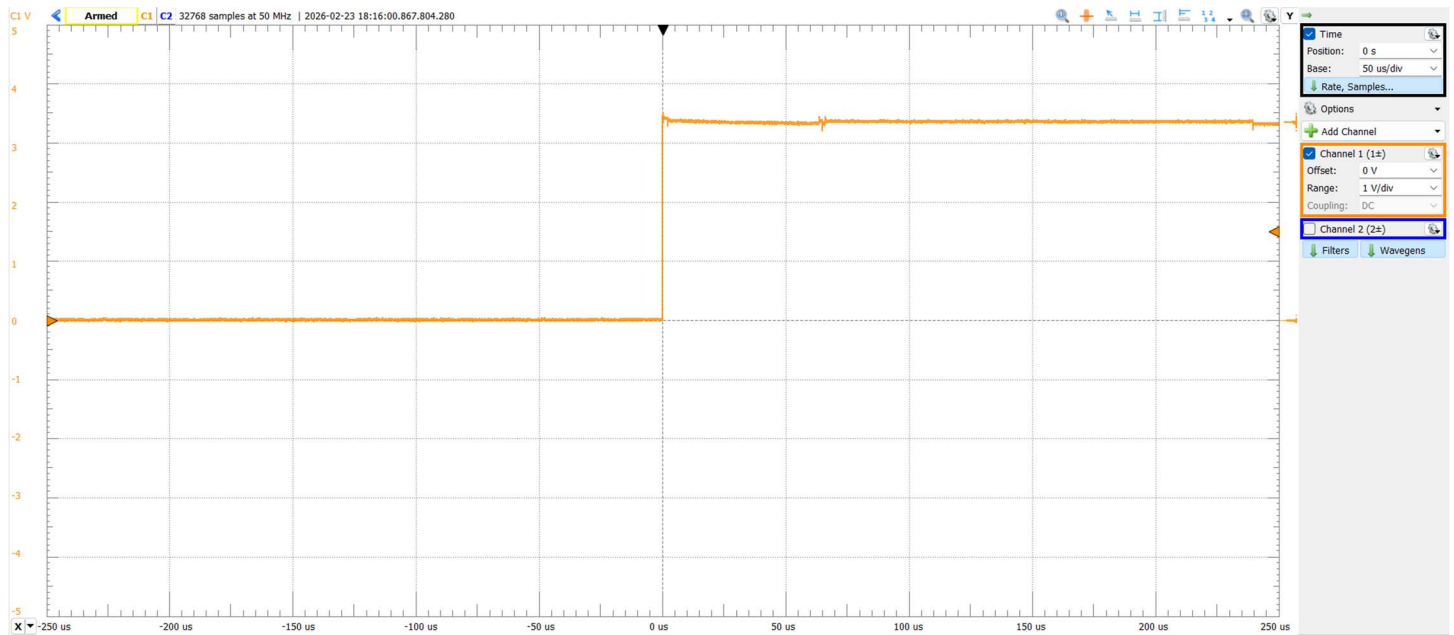


Figure 6: Debounced Switch Response 2

The debounced circuit shows more consistency, and because it never drops past the threshold, the clock never sees a rising edge outside of the initial switching behavior.

Table 1: Next-state truth table for 2-bit Counter

Q1	Q0	Q1+	Q0+
0	0	0	1
0	1	1	1
1	1	1	0
1	0	0	0

Table 2: D-flip flop $D_i = Q_i^+$ truth table

Q1	Q0	D1	D0
0	0	0	1
0	1	1	1
1	1	1	0
1	0	0	0

MPOS(D1):

$$(Q0+Q1)*(Q0+/Q1)$$

MSOP(D1):

$$(Q0*Q1)+(Q0*/Q1)$$

MPOS(D0):

$$(/Q0+/Q1)*(/Q0+Q1)$$

MSOP(D0):

$$(/Q0*/Q1)+(/Q0*Q1)$$

D0:

Table 3: K-Map for D0

Q1\Q0	0	1
0	1	1
1	0	0

$D0 = \neg Q1$

D1:

Table 4: K-Map for D1

Q1\Q0	0	1
0		1
1		1

$D1 = Q0$

Lab 3, Clocked 7-Segment Decoder
Name: Evan Baesler
Class #: 11081
PI Name: Tyler Ruble
Description: Generating outputs 0 thru 3 on the decoder from lab2 utilizing switches for clock and toggle.

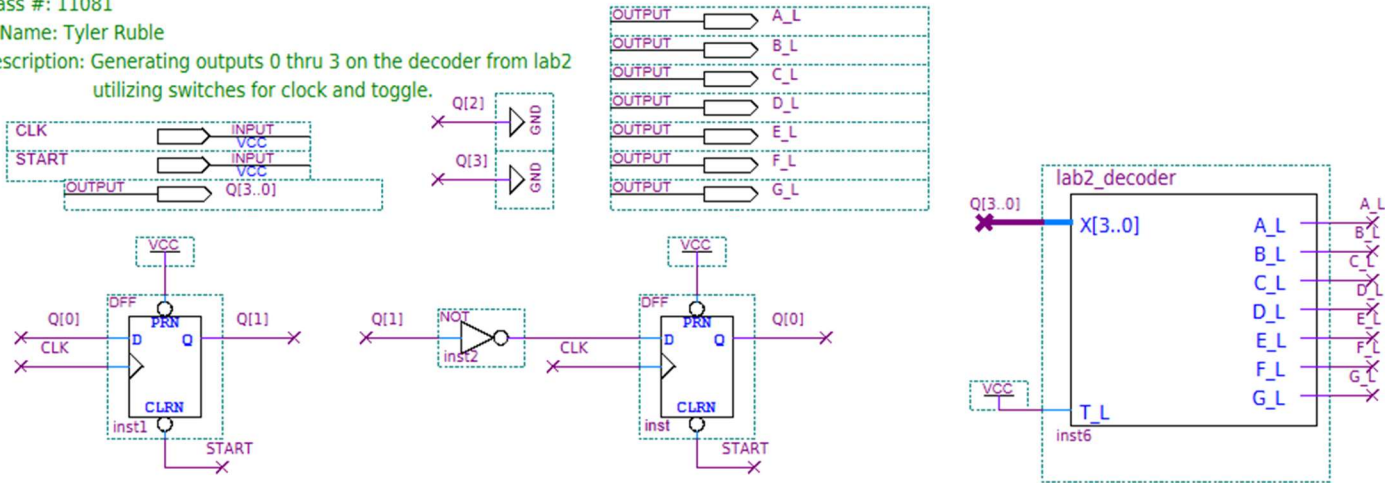


Figure 7: Two-bit counter Quartus BDF

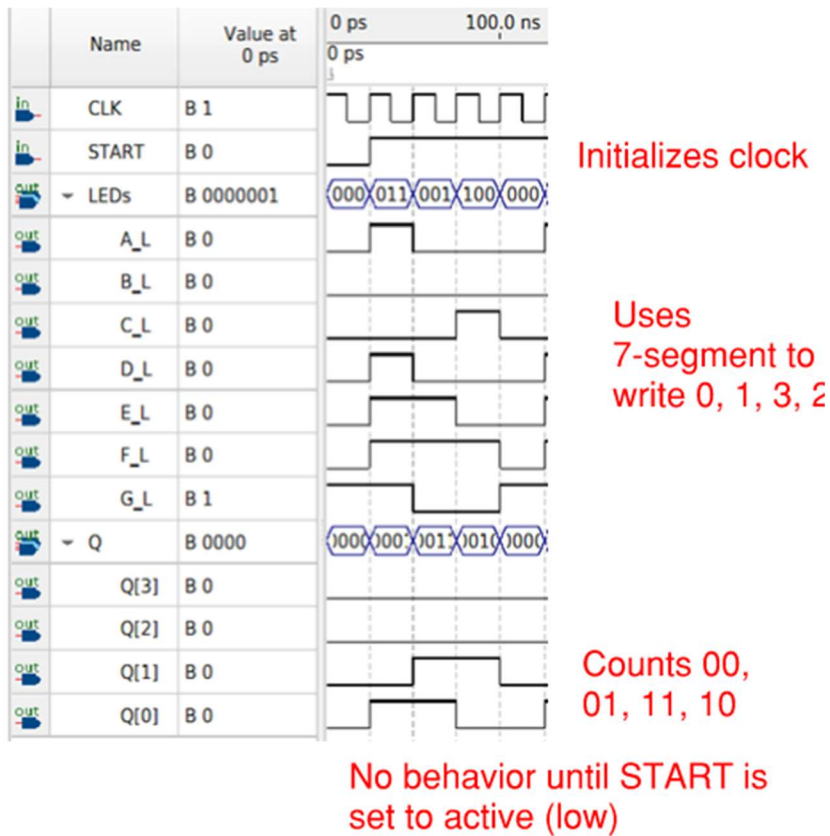


Figure 8: Annotated Quartus Waveform for 2-bit Counter

Outputs: 0,1,3,2,0,1,3,2,0,1

These outputs match the expected and show a working debounce.

Table 5: Truth Table for 3-bit Counter with Forward & Back

F	B	Q2	Q1	Q0	Q2 ⁺	Q1 ⁺	Q0 ⁺	Sp.
0	0	0	0	0	0	0	0	0
0	0	0	0	1	-	-	-	-
0	0	0	1	0	0	1	0	1
0	0	0	1	1	0	1	1	0
0	0	1	0	0	1	0	0	0
0	0	1	0	1	-	-	-	-
0	0	1	1	0	-	-	-	-
0	0	1	1	1	1	1	1	1
0	1	0	0	0	0	1	1	0
0	1	0	0	1	-	-	-	-
0	1	0	1	0	1	1	1	1
0	1	0	1	1	1	0	0	0
0	1	1	0	0	0	1	0	0
0	1	1	0	1	-	-	-	-
0	1	1	1	0	-	-	-	-
0	1	1	1	1	0	0	0	1
1	0	0	0	0	1	1	1	0
1	0	0	0	1	-	-	-	-
1	0	0	1	0	1	0	0	0
1	0	0	1	1	0	0	0	0
1	0	1	0	0	0	1	1	0
1	0	1	0	1	-	-	-	-
1	0	1	1	0	-	-	-	-
1	0	1	1	1	0	1	0	1

$$D0 = Q0^+, D1 = Q1^+, D2 = Q2^+$$

Bolded Terms are Do Not Cares (-) turned into 1 or 0

For $B = 0$, $D0$:

Table 6: $D0$ K-Map for $B = 0$

$F*Q2 \setminus Q1*Q0$	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	1	1	0	0
10	1	1	0	0

For $B = 1$, $D0$:

Table 7: $D0$ K-Map for $B = 1$

$F*Q2 \setminus Q1*Q0$	00	01	11	10
00	1	0	0	1
01	0	0	0	0
11	0	0	0	0
10	0	0	0	0

K-Map Simplification (MSOP):

$$D0 = (\neg B * F * Q1) + (\neg B * \neg F * Q1 * Q0) + (B * \neg F * \neg Q2 * \neg Q0)$$

$$D0 = (F * \neg Q1) + (\neg F * \neg B * Q0) + (B * \neg Q2 * \neg Q0)$$

For $B = 0$, $D1$:

Table 8: $D1$ K-Map for $B = 0$

$F*Q2 \setminus Q1*Q0$	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	1	1	1	1
10	1	1	0	0

For $B = 1$, $D1$:

Table 9: $D1$ K-Map for $B = 1$

$F*Q2 \setminus Q1*Q0$	00	01	11	10
00	1	1	0	1
01	1	1	0	1
11	1	1	0	1
10	1	1	0	1

K-Map Simplification (MSOP):

$$D1 = (\neg B \wedge \neg F \wedge Q1) + (\neg B \wedge F \wedge \neg Q1) + (\neg B \wedge F \wedge Q2) + (B \wedge \neg Q1) + (B \wedge Q1 \wedge \neg Q0)$$

$$D1 = (B \wedge \neg Q0) + (F \wedge \neg Q1) + (F \wedge Q2) + (\neg F \wedge B \wedge Q1)$$

For $B = 0$, $D2$:

Table 10: $D2$ K-Map for $B = 0$

$F*Q2 \setminus Q1*Q0$	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	0	0	0	0
10	1	0	0	1

For $B = 1$, $D2$:

Table 11: $D2$ K-Map for $B = 1$

$F*Q2 \setminus Q1*Q0$	00	01	11	10
00	0	0	1	1
01	0	0	0	0
11	0	0	0	0
10	0	0	0	0

K-Map Simplification (MSOP):

$$D2 = (\neg B \wedge \neg F \wedge Q2) + (\neg B \wedge F \wedge \neg Q2 \wedge Q0) + (B \wedge \neg F \wedge Q2 \wedge Q1)$$

$$D2 = (\neg F \wedge B \wedge Q2) + (B \wedge Q2 \wedge Q1) + (F \wedge \neg Q2 \wedge \neg Q0)$$

For $B = 0$, S_p :

Table 12: S_p K-Map for $B = 0$

$F*Q2 \setminus Q1*Q0$	00	01	11	10
00	0	0	0	1
01	0	0	1	1
11	0	0	1	1
10	0	0	0	0

For $B = 1$, S_p :

Table 13: S_p K-Map for $B = 1$

$F*Q2 \setminus Q1*Q0$	00	01	11	10
00	0	0	0	1
01	0	1	1	1
11	0	0	0	1
10	0	0	0	1

K-Map Simplification (MSOP):

$$S_p = (/B*Q2*Q0) + (/F*/B*Q1*/Q0) + (B*/F*Q2*Q0) + (B*Q1*/Q0)$$

$$S_p = (Q2*Q0) + (/F*Q1*/Q0)$$

Summary:

$$D0 = (F*/Q1) + (/F*/B*Q0) + (B*/Q2*/Q0)$$

$$D1 = (B*/Q0) + (F*/Q1) + (F*Q2) + (/F*/B*Q1)$$

$$D2 = (/F*/B*Q2) + (B*/Q2*Q1) + (F*/Q2*/Q0)$$

$$S_p = (Q2*Q0) + (/F*Q1*/Q0)$$

Figure 10: Annotated Quartus Waveform Simulation for 3-bit Counter

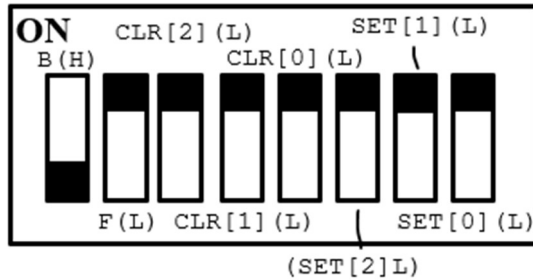


Figure 11: Switch Legend for 3-bit counter

Next States for 001:

F	B	Next State
0	0	001
0	1	000
1	0	110
1	1	110

Next States for 101:

F	B	Next State
0	0	111
0	1	011
1	0	010
1	1	010

Next States for 110:

F	B	Next State
0	0	110
0	1	010
1	0	010
1	1	010

Q1	Q0	T1	J0	K0	Q1+	Q0+
0	0	0	1	-	0	1
0	1	1	-	0	1	1
1	1	0		1	1	0
1	0	1	0	-	0	0

Q1	Q0	J0	K0	Q0+
0	0	1	-	1
0	1	-	0	1
1	1	-	1	0
1	0	0	-	0

$$J0 = (/Q1)$$

$$K0 = (Q1)$$

$$Q0+ = (/Q0 * J0) + (Q0 * /K0)$$

Q1	Q0	T1	Q1+
0	0	0	0
0	1	1	1
1	1	0	1
1	0	1	0

$$T = Q1 \text{ XOR } Q0$$

$$Q1+ = (/Q1 * T1) + (Q1 * /T1) \rightarrow Q1 \text{ XOR } T1$$

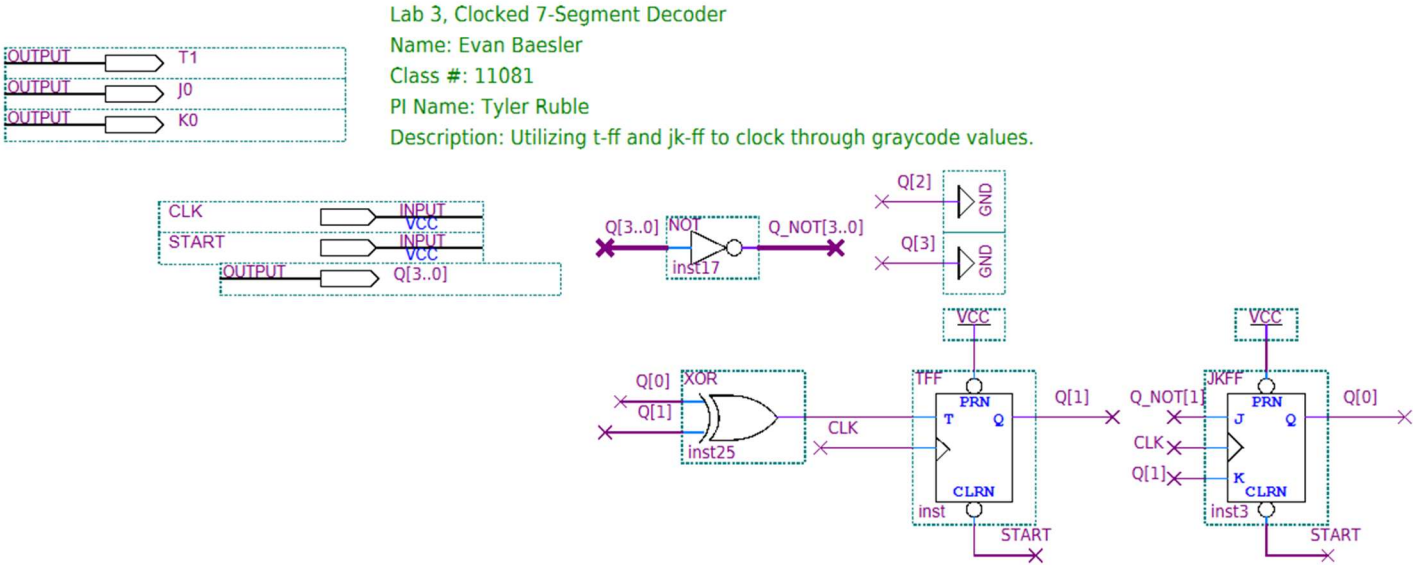


Figure 12: Quartus BDF for tff and jkff counter

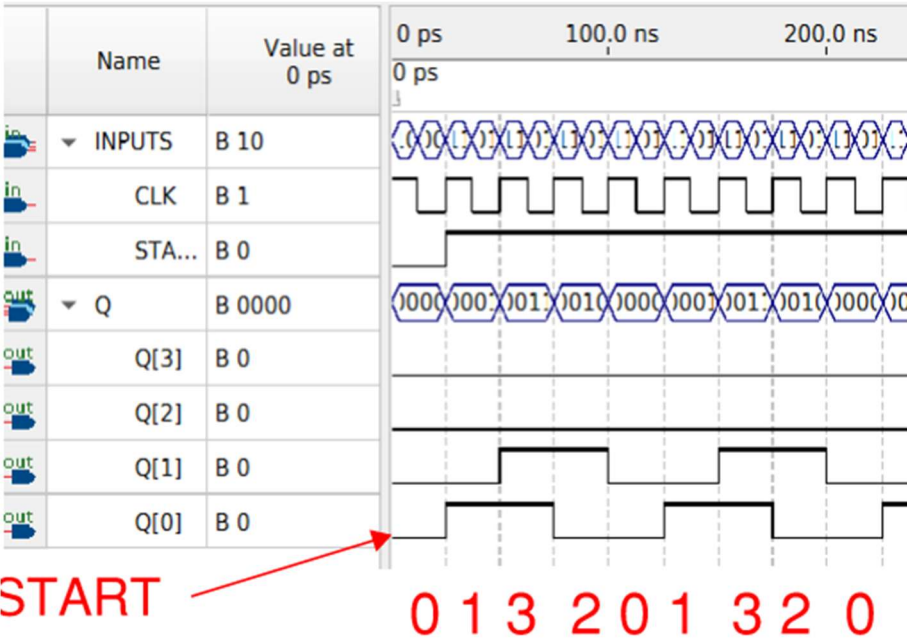


Figure 13: Annotated Quartus Waveform for tff and jkff counter